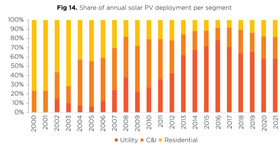
solar

solar pv

on grid

- utility — large-scale installations through tenders and auctions, allowing a rapid and cost-efficient development of large solar capacities. It is simpler than creating a high-volume rooftop solar market that requires a functioning legislative framework and create awareness for solar among citizens and businesses.
- rooftop — As the systems can be attached to or integrated into buildings, it avoids any conflicts related to additional land use for solar system deployment. Moreover, being installed very close to the point of consumption, rooftop solar offers the benefit of reducing grid costs and developing a decentralised energy system based on prosumers.
  - category
    - residential
    - commercial & industrial (C&I)
- deployment

Fig 14. Share of annual solar PV deployment per segment



Source: IEA, Bloomberg, IHS

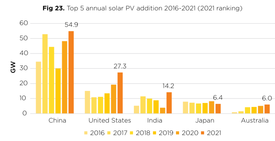
Fig 13. Annual global solar PV deployment per segment, 2000-2021

Source: IEA, Bloomberg, IHS

Notes: The reason for residential solar PV' s exceptional performance in recent years can be attributed to China, which introduced a favorable policy scheme (leading to installation of around 21 GW of residential PV systems in 2021, equal to 70% of the entire segment), but at early stage, solar capacity growth in China was mainly from utility-scale. Recent policy shift may aim at absorbing overcapacity in solar industry while there is limited room for further utility expansion

- Key countries

Fig 23. Top 5 annual solar PV addition 2016-2021 (2021 ranking)



Source: SPE

others

- building intergrated pv
  - Desc: these are products that are integrated directly into the building envelope—such as the facade, roof, windows, or skylights—serving both as a building material and as a power generator.
  - Deployment: still niche market, mainly promoted by European countries but slowed down as the targeted policy stopped
- floating pv
  - Desc: solar power systems installed on bodies of water, such as lakes, reservoirs, ponds, or even the sea. It has benefits for not competing for land with agricultural or industrial activity
  - Deployment

Fig 33. Floating-PV annual and total installation capacity

Source: IEA-PVPS (2021)

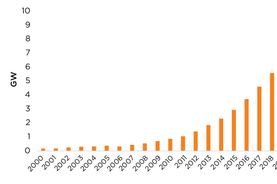
  - total installation capacity is still low, but keeps growing
  - Key countries
    - Majority of installtion is located in **Eastern Asia**. **Mountainous countries such as Japan and South Korea** are looking seriously in floating PV as a solution for their land-scarcity issue which limit the ground-mounted utility-scale deployment. The city-state of **Singapore**, also affected by **land scarcity**, opened a 60 MW floating plant in 2021 on its main water reservoir.
    - Mountainous countries with limited access to the sea can also benefits from floating PV: **Switzerland** turned on an installation in **a high-altitude reservoir**. The high-altitude conditions increase the irradiation levels and increase in turn the actual production of solar panels.
    - **Indonesia** is also active in the segment and is currently building a 145 MW system on the Cirata Reservoir in West Java.
    - The current **largest floating PV also belongs to China**, with 320 MW located in Dezhou. A major advantage of floating PV is the possibility to **combine it with existing hydropower dam.**
- Agri-pv
  - Desc: a dual-use land approach that solar panels are installed at a height, spacing, or orientation that lets sunlight, rain, and equipment reach the crops below. It has benefits that reduce land competition between solar and agriculture while guaranteeing the efficiency and viability of both activities.
  - key countries
    - **Japan, South Korea, and China**(China is also home of the largest agri-PV project, with a capacity of 1 GW in the Ningxia Province)
    - Other countries such as **Chile, the Gambia, and Mali** are equipped with major installations.
    - In **Europe**, the sector's growth is mainly concentrated in France, Greece, the Netherlands and Spain.

off grid

Desc: Off-grid systems: solar systems that provide electricity to households and villages that are not connected to the utility electricity network. The benefit is low-skill-requirement with "plug-and-play" nature (see examples in my first draft of report) as well as affordability

- Category
  - lighting products (e.g., solar lantern)
  - solar home systems — Systems up to 20W are used to power several lights, mobiles power stations and a radio. As the capacity grows, systems can include a television, fans or refrigerators
  - ...
- Deployment

Fig 35. Cumulative off-grid solar PV capacity 2000-2021



Source: IEA, IRENA, IHS

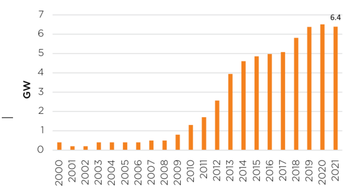
Notes: Cumulative installation is still limited, constituting less than 1% of total PV capacity. As still over 700 million people miss access to stable electric supply, off-grid PV remains a large market with nearly 500 million users worldwide in 2021.

- Key countries
  - Most off-grid installations are founded in **Asia-Pacific and in Sub-Saharan Africa (SSA).**
    - SSA countries can rely on different options to increase their access to electricity but geographic characteristics can impact their choices. Grid-extension is easier for smaller countries such as Rwanda, Burundi or Uganda while **larger countries such as Angola, Democratic Republic of Congo or Niger will rely more on SHS systems and mini-grids**. In Asia Pacific, the market is more developed, and off-grid mostly serves as a grid extension. For **islands in the Pacific**, which are very dependent on expensive fossil fuel imports, off grid solar allow to **reduce their import expenses and turn to greener solution** at the same time.
    - **Kenya** is the largest and most mature market for solar off-grid solutions globally, and accounts for almost 74% of solar home system sales in East Africa in 2023. (IEA Kenya, 2024). Through several policies, the government has **simplified the access to financing opportunities** and **removed legislative barriers for self-consumption**, boosting the installation of solar home systems. Kenya is also a pioneer in the development of **mini-grid projects**. It first started to allow small projects to be run with minimal oversight, and it eventually turned into larger programmes such as the Kenya Off-Grid Solar Project, aiming at empowering the least developed part of the country.
  - There are also developed countries with large proportion of off-grid solar — Finland' s relatively high proportion of off-grid solar systems reflects not a weak grid, but a combination of geographic spread, high grid connection costs in remote areas, strong cultural values of self-sufficiency, and modern solar+storage technology that makes off-grid living viable even in northern climates.

solar thermal electricity

- Concentrated Solar Power(CSP)
  - Desc: a solar thermal technology that uses mirrors or lenses to concentrate sunlight into heat, which is then used to produce electricity—unlike PV, it generates power through thermal energy rather than the photovoltaic effect. It is suitable for regions with good Direct Normal Irradiation (DNI) resource like Africa
  - Category
    - Linear Fresnel reflector
    - Parabolic trough (solar)
    - Solar tower
  - Deployment

Fig 41. Total CSP installation



Source: IRENA, IEA

Notes: In the early days CSP is considered a very attractive way to harnessing solar for power generation, but has not fulfilled the promises, only increasing total installed global capacities from 0.4 GW in 2000, to 1.3 GW in 2010, and 6.4 GW by 2021

  - Key countries

Source: https://cspdata.com/

Notes: Ambitious projects (e.g., Desertec Initiative) in North Africa proposed massive CSP installationsare in Sahara Desert, not only to meet local energy demands but also to export renewable power to Europe. However, CSP faced significant practical and logistical challenges. Firstly, CSP plants are typically located in arid desert regions far from major population centers, requiring **extensive and costly cross-border transmission infrastructure**. The Desertec plan, for example, **involved complex transnational coordination and enormous investments in grid interconnections across multiple countries**, which ultimately proved politically and economically infeasible. Additionally, CSP technology **demands substantial quantities of water for cooling and mirror cleaning**—resources that are particularly scarce in desert environments.

- Source — made by myself and mainly based on IEA data explorer as well as ISA world solar power report(2022). Please feel free to point out if there's anything unclear or in need of more details!